A Free-Scale First-Root Carrier
Fifteen Integers, Degree 44, 111 Gates

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1 A free-scale carrier of degree 44

This endpoint uses fifteen unrestricted integer variables, has exact degree 44, and a division-free evaluator with 111 binary arithmetic gates (63 multiplications, 26 additions, 22 subtractions). It improves the prior degree 48 at a cost of three gates; it does not Pareto-dominate the retained 108-gate circuit. The same elementary projection factorization reduces that old evaluator to 107 gates, so the apples-to-apples evaluation cost is four gates. The new device is to release the scale variable, replace premature coprimality by a magnitude guard, and recover the exact scale only after an amplified whole-product projection.

1.1 Complete polynomial

The variables are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_1, p_2, p_3 \in \mathbb{Z}.$$

The fixed numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are exactly the six pinned source34 coefficient payloads; M_F is already shifted. All the following are polynomial abbreviations, not witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = 2X^2 - 3Xc - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_0 = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
B_{\mathrm{new}} = Y^2B_0 - k^2q^8, \qquad P_{\mathrm{os}} = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{gathered}$$

Define the residuals

$$\begin{aligned}
A &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_{\mathrm{aux}} &= V^2 - (Q - 1)y(2V + y) - 1, \\
G &= B_{\mathrm{new}} - P_{\mathrm{os}}.
\end{aligned}$$

The evaluator uses the equivalent schedules

$$K_{\mathrm{lo}} = X(2X - 3c) - 2, \qquad B_{\mathrm{new}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big(4(Y\eta)^2 + (kq^3)^2\big),$$

reusing Yk and q^3. The sole equation is

$$\boxed{P = A^2 + B^2 + C_{\mathrm{aux}}^2 + G^2 = 0.}$$

1.2 Monicity before coprimality

The actual $\beta = 2^{1064921} - 1$ is odd and 1 (mod 3). Consequently q is nonzero, even, and 2 (mod 3). The polynomial H is odd and nonzero, so $U \neq 0$ on every assignment. At a zero, $t = Z/U \in \mathbb{Q}$ satisfies

$$t^2 + 2(2L + k)t + k^2 - 4 = 0.$$

Monicity makes t integral. Modulo 2, $t + k$ is even, so $s = (t + k)/2$ is integral and

$$s^2 + (2s - k)L - 1 = 0, \qquad (s + L)^2 = 1 + L(L + k).$$

The same divisibility $U \mid Z$ gives

$$M \mid H(Y - q^3), \qquad H \mid MK_{\mathrm{lo}}.$$

There is deliberately no early assertion that H and q are coprime. Since H is odd and q even, the first relation nevertheless gives Y even, hence Δ odd. If $K = X(X + 2ac) - 1$, then $2K - K_{\mathrm{lo}} = HXc$, so oddness of H gives

$$H \mid q^3K.$$

1.3 The guard replaces the early scale hypothesis

The polynomial P_{os} is strictly positive and represents every positive integer by the three-square theorem. Hence

$$Y^2B_0 > k^2q^8 > 0,$$

where $k \neq 0$ follows because $k = 0$ would make $B_{\mathrm{new}} \leq 0$. Since $B_0 \leq k^2q^2$, this proves

$$B_0 > 0, \qquad |Y| > |q|^3.$$

The unchanged centered-budget argument gives common strict signs for $q, F, C, \alpha = q - F - C$, and

$$|\eta/k| < \tfrac{1}{2}, \qquad |k|(|Y| - \tfrac{1}{2}) < |c| < |k|(|Y| + \tfrac{1}{2}), \qquad \operatorname{sgn}(c) = \operatorname{sgn}(kY).$$

Put $q_* = |q|$, $x = |X|$, $m = |Y|$. The authentic packing estimates give

$$q_* \geq 16, \quad q_*(q_*^2 - 1) < R < q_*^4 - 2, \quad x \geq 4q_* \geq 64, \quad |E| > 4q_*^4 > 3R + 1.$$

These replace the old literal cubic-scale expression wherever it was used before projection. Its remaining parity consequence, Δ odd, was already proved without coprimality.

1.4 Signed Pell bootstrap: inherited statements and hypotheses

The unchanged strong and auxiliary residuals recover the missing main Pell ordinate by the predecessor's odd-index folding argument. For $B_p = |a+2|$, $c_* = |c|$, and $I_* = |2r+1|c_*^2$, they give

$$|f|^2 - (B_p^2 - 1)I_*^2 = 1, \qquad |f| > 2I_*.$$

With $S = \Delta I_*$ the auxiliary row is

$$S^2V^2 - (S^2-1)(V+y)^2 = 1.$$

The auxiliary Pell index is odd. Its polynomial identity, reduced modulo $|f|$, gives an ordinate congruent to $\pm c_*$. Folding the index into the interval from 0 to the strong index makes the ordinate at most I_*. Since $I_* + c_* < |f|$, this congruence gives an actual main ordinate $c_* = \psi_{B_p}(p)$.

The first Pell classification gives $|k| = 2\psi_{|2XY^2+1|}(n_0)$. The inequality $|c| > |k|$, the parameter comparison, the congruence $k = R + 1 + hE$, and $|E| > 3R+1$ give the same signed first-index lower bound as in the source. The unchanged strong-rank lemma and signed auxiliary step-down then give

$$p = R, \qquad 2n_0 = R+1, \qquad R = 2s_0 + 1, \quad s_0 \geq 24.$$

The sign-uniform ratio estimate uses only these classifications, $x \geq 64$, $m \geq 6$, and $|c/k| < m+1$. It proves

$$m > \frac{(x-3)^{s_0}}{2} - 1, \qquad |H| > x(x-3)^{s_0}.$$

These are precisely the inherited unbounded Pell steps, not conclusions of finite testing. Their complete statements and the verified hypotheses are detailed in `PROOF.md`, Sections 3–4, with pinned source texts. No main projection or power-of-two property has yet been used.

1.5 Amplified projection and late scale recovery

Let $\sigma = \operatorname{sgn}(c)$ and $x_0 = \sigma X$. The signed polynomial Pell projections give

$$2^R K \equiv (x_0 - 2^R)(2^R x_0 + 1) \pmod{H}.$$

Thus H divides q^3 times this whole product. We bound that larger product instead of canceling a possible nonunit modulo H. For $x \geq 64$ and $s_0 \geq 24$,

$$(x/32)^{s_0} \geq (x/32)^{24} \geq x^4,$$

since $x^{20} \geq 64^{20} = 32^{24}$. Therefore

$$|H| > x(x-3)^{s_0} \geq x(x/2)^{s_0} \geq 16^{s_0}x^5.$$

Since $q_* \leq x/4$ and $R = 2s_0 + 1$,

$$q_*^3 2^{2R+2} x^2 \leq \frac{x^3}{64} 16^{s_0+1} x^2 = \tfrac{1}{4} 16^{s_0} x^5 < |H|.$$

This strictly bounds the absolute value of the divisible whole product. The product is zero. Its factor $2^R x_0 + 1$ is odd and cannot vanish, so $x_0 = 2^R$. Because $X \equiv 2 \pmod 3$ and R is odd, $\sigma = 1$ and $X = 2^R$.

The zero whole product and the same Pell congruence now give $H \mid 2^R K$. Since H is odd, $H \mid K$ and hence $H \mid K_{\mathrm{lo}}$. Set $g = K_{\mathrm{lo}}/H \in \mathbb{Z}$. Substituting $K_{\mathrm{lo}} = Hg$ into $MH \mid Z$ and canceling the nonzero integer H gives

$$M \mid Y - q^3, \qquad v_{\mathrm{scale}} = (Y - q^3)/(2q^3) \in \mathbb{Z}.$$

This is cancellation in the integers, not cancellation of a nonunit modulo H. In particular, no CRT coprimality claim is needed. Thus the exact old cubic scale is recovered after, rather than before, the signed bootstrap. The surviving index sign gives $k > 0$; with $c > 0$, this also gives $Y > 0$. The inherited centered half-binomial extraction now applies without change.

1.6 Explicit maps and canonical completeness

From any new zero construct a degree-48 predecessor zero as follows. Retain $j, C, z, h, r, w, T, c, f, y$, and set

$$v_{\mathrm{old}} = Y/q, \qquad n_{\mathrm{old}} = t - g - v_{\mathrm{scale}}.$$

The old carrier's quotient is exactly

$$\frac{Z_{\mathrm{old}}}{U_{\mathrm{old}}} = n_{\mathrm{old}} + g + \frac{v_{\mathrm{old}} - q^2}{2q^2} = t.$$

Its strong and auxiliary rows are unchanged. Represent the positive integer B_0 by its three Pos coordinates. This restores a literal old zero, so the old native/compiler/refutation theorem applies.

Conversely start with a finite syntactic refutation of ZFC and use the same authenticated canonical compiler family. It supplies an old zero with $q > 0$, $X = 2^R$, $s_0 \geq 24$, the exact cubic scale, and

$$D_{\mathrm{err}} \leq q^2/2, \quad 0 < \eta/k < 1/4, \quad B_0 > q^2k^2/4, \quad Y \geq X^{s_0}/2 > 2q^3.$$

Consequently

$$B_{\mathrm{new}} > k^2q^2(Y^2/4 - q^6) > 0.$$

Retain the arithmetic coordinates, set $v = Y$ and $n = t - g - v_{\mathrm{scale}}$, and represent B_{new} by the three Pos coordinates. Then $Z/U = t$, and all four new rows vanish. These constructions prove equivalence of zero existence; a bijection of arbitrary zero sets is not asserted.

1.7 Exact degree, reproducibility, and boundary

The four residual degrees are $22, 12, 16, 18$. In degree 22, the first residual contributes only $4ULZ$. With subscript 0 for leading homogeneous forms,

$$q_0 = 6\beta j, \quad X_0 = 12q_0w, \quad Y_0 = v,$$
$$R_0 = -q_0^3(12wC - q_0z), \quad U_0 = 8q_0^3X_0Y_0, \quad L_0 = X_0Y_0^2R_0,$$
$$Z_0 = U_0n + 4q_0^3X_0^2, \qquad P_{44} = 16U_0^2L_0^2Z_0^2.$$

In particular,

$$[j^{26}w^6v^8z^2n^2]P = 2^{54}3^{32}\beta^{26} > 0.$$

The exponents sum to 44, and the other squared rows have smaller degree. The exact degree therefore holds after fixing every coefficient.

The accompanying independent cut-register audit authenticates coefficient bytes, checks the literal graph, exact identities and leading coefficient, and rejects altered arithmetic gates. These finite checks validate the transcription and algebra, not the unbounded inherited compiler theorem. The result is an ordinary arithmetic equivalence, hence an equivalence provable in ordinary ZFC between no integer zeros and syntactic Con(ZFC). It is not a proof of consistency, a global optimum, or a new formal-kernel certificate.